

CoMem: Persistent Intermediate-Residual Memory for Bounded-Read Long-Context Inference

Anonymous ACL submission

Abstract

We study repeated queries over stable documents. **CoMem** stores one depth- j residual per token, retrieves a bounded chunk set, and resumes layers $[j:L)$. In a matched comparison with identical evidence, mask, examples, and adapter, $j=12$ reduces isolated model Read from 931.9 to 664.4 ms ($1.403\times$) while lowering a 15-cell RULER macro from 99.19 to 96.07 (paired gap 3.12, 95% CI [2.36, 3.93]). A rank-32 self-distillation adapter makes the interface usable without updating the backbone. The 8 KiB/token store requires reuse: at 32k, CPU-pinned break-even is 8.9–10.9 queries for generations up to 128 tokens; at 128k and one generated token it is 25.8–27.6 across GPU/CPU placement. In a nine-cell equal-latency diagnostic, dependence-aware hierarchical intervals show BM25 replay ahead by 11.56 points ($[-18.67, -5.11]$), while frozen-BGE replay has an unresolved 1.00-point edge ($[-10.67, 8.33]$); CoMem wins neither aggregate. Restoring lower-layer document context raises a paired multikey diagnostic from 92.5 to 100.0, and a 32-token overlap reaches 98.5 without increasing persistent bytes or per-query Read. CoMem is an internal measurement of a depth-reuse operating point, not a claim of superiority over raw-text retrieval or modular-cache systems.

1 Introduction

Repeated-query long-context serving can save work in two different ways. Raw-text retrieval bounds the online token set but recomputes all layers (Lewis et al., 2020; Xu et al., 2024). Prefix, position-independent caching (PIC), and modular-cache systems instead reuse precomputed document state, usually layer-wise KV, while repairing position or context dependencies (Yao et al., 2025; Lu et al., 2025; Hu et al., 2025; Agarwal et al., 2025; Chen et al., 2026). This paper asks

a narrower question: *how much transformer depth should a reusable document object prepay?*

Workload contract. We assume an open-weight model, a shared or slowly changing corpus, and enough queries per document to amortize a model-version-specific Write. One-off queries, frequently edited text, or corpora whose persistent states cannot be kept near the accelerator may favor raw text. Every latency claim below names whether it includes selection, state fetch, Write, model Read, and decode.

We instantiate that question with **CoMem** (Figure 1). **WRITE** runs each context chunk through layers $[0:j)$ and stores one residual vector per token. **SELECT** chooses a fixed number of chunks. **READ** packs their residuals with the query and executes only $[j:L)$ with full cross-pack attention. The endpoint $j=0$ stores token IDs and replays the same selected pack through every layer. Thus j varies prepaid depth while holding evidence fixed.

Our strongest evidence is this *matched $j=0 \rightarrow 12$ operating-point comparison*. On Qwen3-8B, $j=12$ lowers isolated selected-pack Read from 931.9 to 664.4 ms ($1.403\times$) but lowers a paired 15-cell RULER macro from 99.19 to 96.07. This is not quality-preserving acceleration. A self-distilled upper-layer LoRA is necessary to make the residual interface usable, and the 8 KiB/token store is worthwhile only when reused: CPU-pinned break-even ranges from 8.9–10.9 queries at 32k for generations up to 128 tokens, while the full GPU/CPU grid ranges from 5.5–10.9. At 128k we retain only the one-generated-token points, 25.8–27.6 queries by store placement. Much larger dense-context numbers combine bounded selection with depth reuse and are secondary operating points, not the causal depth result.

We also test whether saved lower-layer compute can buy enough extra evidence to offset the residual-interface loss. The answer is selector-

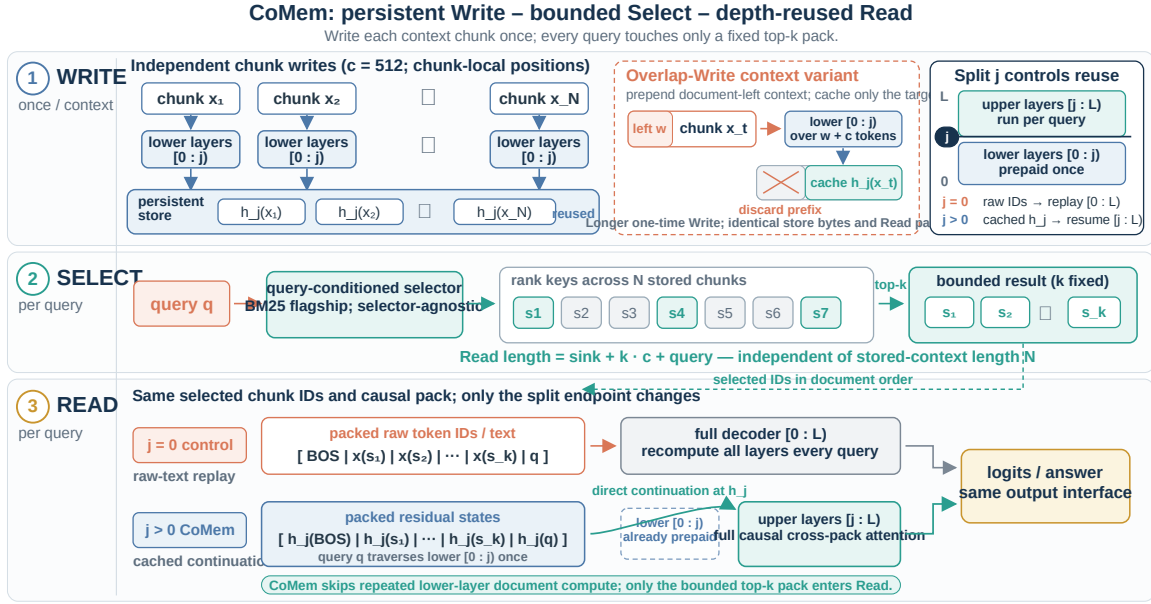


Figure 1: **CoMem treats transformer depth as a cross-query reuse axis.** WRITE stores one intermediate residual per token, SELECT retrieves a fixed working set, and READ resumes the upper layers with the query present. The matched $j=0$ path retrieves the same chunks, order, sink, and mask as raw text but recomputes every layer. The reported $1.403\times$ is this matched depth effect; larger dense-to-bounded speedups also include selection. A small left-overlap is evaluated as a focused repair for missing lower-layer document context.

dependent, but never a CoMem aggregate win in our nine-cell diagnostic mixture. At matched online latency, iterative-BM25 raw replay (maximum budget 10 chunks) beats CoMem (budget 12) 64.78 to 53.22; frozen-BGE replay at budget 10 is 54.22 versus 53.22, with a dependence-aware hierarchical interval that crosses zero. Selectors may under-fill these maximum budgets. Thus this experiment bounds deployment conclusions rather than establishes a new quality ranking over RAG or reusable-KV systems.

We also bound one source of error rather than claiming a complete explanation. On a paired 8k/16k multikey diagnostic, document-contextual lower-layer states close the 92.5-to-100.0 gap, while document-origin position remapping alone reduces accuracy to 88.0. A deployable 32-token left overlap reaches 98.5 with unchanged persistent bytes and per-query Read work. Because this repair is not evaluated on the full natural-task suite, we treat it as a localized diagnosis and engineering hypothesis.

Contributions.

- **Matched depth endpoint:** a same-evidence, same-adaptor measurement of the incremental benefit and quality cost of reusing lower-layer

computation.

- **Transparent system boundary:** persistent bytes, Write amortization, serving crossover, and selector-dependent equal-latency results, with bounded selection kept separate from depth reuse.
- **Bounded diagnosis:** controlled continuous-state, context, position, and overlap variants that isolate a major Write-side failure on one paired multikey cohort without generalizing it beyond the evidence.

2 Related Work

We position CoMem by stored object and post-retrieval computation. Table 1 is an interface map, not an empirical ranking.

PIC and modular document caches. CacheBlend, TurboRAG, RAGCache, and Cache-Craft precompute or reuse retrieved chunk KV and handle context-dependent composition in different ways (Yao et al., 2025; Lu et al., 2025; Jin et al., 2025; Agarwal et al., 2025). PIC systems such as EPIC and MEPIC splice independently encoded per-layer KV and repair boundaries or selected tokens; APE uses parallel

Family	Persistent object	Composition / repair	Post-selection model work	Relation to CoMem
Text RAG (Lewis et al., 2020)	token IDs	concatenate retrieved text	all layers	matched $j=0$ endpoint
Chunk-KV reuse (Yao et al., 2025; Lu et al., 2025; Agarwal et al., 2025)	per-layer KV	fuse / partially recompute	repair or decode	closest serving workload
PIC / parallel encoding (Hu et al., 2025; Wang et al., 2025; Yang et al., 2025b)	per-layer KV	position/boundary repair	selected repair or decode	independent modular chunks
Learned modular KV (Chen et al., 2026; Eyuboglu et al., 2026; Xie et al., 2026)	learned per-layer KV	trained compiler/adapters	little/no document recompute	closest learned objects
Activation replay (Gao et al., 2025; Qasim et al., 2026)	activation or residual	restore/reconstruct state	replay suffix or KV	intermediate-state precedent
CoMem	one depth- j residual/token	pack selected residuals	execute layers $[j:L)$	measured depth-reuse endpoint

Table 1: **Positioning among reusable-context interfaces.** This taxonomy does not rank systems: the cited methods differ in backbone, training, workload, hardware, storage tier, and timing boundary. CoMem’s specific object is one residual per token at a tunable depth; the empirical claim in this paper is the matched $j=0$ versus $j=12$ endpoint, not superiority over these families.

context encoding and attention realignment (Hu et al., 2025; Wang et al., 2025; Yang et al., 2025b). Fusion RAG Cache studies selection and cache fusion at serving scale (Wang et al., 2026). Learned modular objects further include KV Packet’s independently compiled KV plus boundary adapters and Cartridges’ distilled reusable KV representations (Chen et al., 2026; Eyuboglu et al., 2026). Concurrent Cartridges at Scale and SemPIC retrieve learned per-document KV objects (Hardalov et al., 2026; Xie et al., 2026). These works already establish reusable document caches. CoMem does *not* claim that idea; it studies the distinct single-residual object, whose split depth trades persistent bytes and online upper-layer computation. We lack a matched same-backbone implementation of these systems, so Table 1 supports taxonomy rather than superiority. No artifact in this study supplies a same-backbone, same-hardware PIC, chunk-KV-repair, or learned modular-cache result under our pack, storage tier, and timing boundary.

Intermediate-state reuse and token reduction.

ReadOnce Transformers and Embedding Recycling cache intermediate text representations and adapt later layers (Lin et al., 2021; Saad-Falcon et al., 2023). HCache checkpoints activations to restore evicted state; KV-Direct reconstructs layer-wise KV from residuals; ILRe and REFORM select or gather tokens before recomputation (Gao et al., 2025; Qasim et al., 2026; Liang et al., 2025; Song et al., 2025). GemFilter likewise selects with early layers but does not persist the states across queries (Shi et al., 2026). Token/KV compression instead changes the retained token set or state budget after full or partial processing (Xiao et al., 2024b; Zhang et al., 2023; Li et al., 2024; Cai et al., 2025; Liu et al., 2024). CoMem’s narrow technical object is one persistent depth- j residual per selected token followed by direct execution of $[j:L)$.

External memory and the matched endpoint.

MemoryLLM, LongMem, and XC-Cache add latent pools or auxiliary readers (Wang et al., 2024, 2023; Monteiro et al., 2024); they are useful external references but not matched depth controls. Our $j=0$ endpoint is deliberately narrower than “RAG”: it fixes selector, chunk IDs and order, sink, mask, examples, and adapter, then replaces stored residuals with token IDs. The paper’s causal claim is therefore about depth reuse under identical evidence, not a ranking over retrieval or memory systems.

Scope of the contribution. Caching intermediate representations and reusable document state both predate CoMem. Our empirical contribution is narrower: one residual per token at a chosen split, direct suffix execution on a bounded selected pack, and a controlled $j=0$ endpoint with explicit Write/Read/storage accounting. We therefore position the paper as an internal measurement and diagnosis of this design point, pending matched comparison with the nearest reusable-context systems.

3 Why Reuse Model Depth?

The interface does not require a layer at which a model has “understood” the document. It requires only that intermediate states retain useful information and that later layers can be adapted to read states produced without the query. Figure 2 supplies motivation, while the matched experiment in Section 5 supplies the actual evidence.

A frozen interface deteriorates as the split moves deeper. With separately distilled adapters, RULER changes from 98.29 at $j=6$ to 96.07 at $j=12$ and 55.41 at $j=18$, while fixed-pack Read speedup rises from $1.166\times$ to $1.403\times$ and $1.807\times$ (Appendix Table 10). Because each split has a different adapter span and parameter count, the retained $j=12$ Write value is missing, and the points are not one compute-matched causal frontier. We

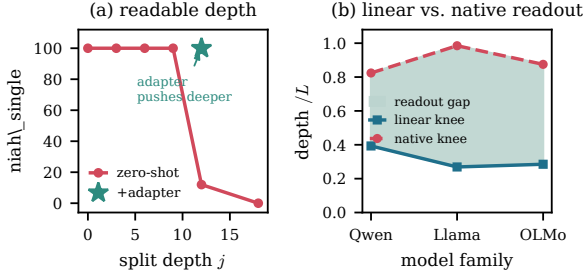


Figure 2: **Intermediate accessibility motivates, but does not validate, depth reuse.** Frozen probes become informative before native LM-head readout, but both measures are protocol-dependent; definitions and controls are in Appendix A.1.

call this a *multi-depth deployment curve*; the paper’s matched causal claim is only the $j=0$ versus $j=12$ endpoint.

Selection and representation also fail separately. BM25 retains lexical single-needle evidence, but on LongEval the distilled $j=12$ reader reaches only 64–76% accuracy conditional on a retrieval hit; BABILong exhibits both misses and hit-conditional readout failures. We therefore evaluate depth under a fixed evidence pack and diagnose Write context only after retrieval is held constant.

4 CoMem

Interface. Let a decoder have L layers and width d , with half-open layer ranges and h_j denoting the input to block j . WRITE divides a document into c -token chunks, runs each through $[0:j)$ with positions restarted at zero, and stores $h_j \in \mathbb{R}^{c \times d}$. SELECT returns top- k chunk IDs. READ orders them by document position, assigns fresh contiguous positions to $[\text{BOS}; \{h_j\}; h_j(q)]$, and runs $[j:L)$ with full causal cross-pack attention. For $j=0$, the same path stores token IDs and executes all layers, giving the matched endpoint of the depth axis.

Storage and bounded model work. With head width d_{head} , query-head count n_q , and KV-head count n_{kv} , residual-to-full-KV bytes/token at a common dtype are

$$\frac{|h_j|}{|\text{KV}|} = \frac{d}{2Ln_{kv}d_{\text{head}}} = \frac{n_q}{2Ln_{kv}}, \quad (1)$$

using $d = n_q d_{\text{head}}$. For Qwen3-8B this is 1/18: 8 KiB rather than 144 KiB/token in bf16. It is about 1 GiB at 128k tokens—far larger than text, so reuse is essential. The model-side Read length

is $\text{sink} + kc + \text{query}$ (nominally 6,657 tokens for $k=12, c=512$), independent of stored-context length. The persistent store, index, and selector are not bounded.

Overlap-Write. Chunk-local writes omit lower-layer left context. For the diagnostic variant, chunk i is prefixed by its preceding w tokens during Write, but only the current c residuals are stored. Thus persistent bytes, selected-pack length, and per-query Read/decode are unchanged; Write work and edit invalidation grow. This is a tested repair on one diagnostic, not the default for the full suite.

Self-distillation. Following knowledge distillation and low-rank adaptation (Hinton et al., 2015; Hu et al., 2022), a $j=12$ student matches a frozen $j=0$ teacher on the teacher’s top-64 logit support at every query token. The teacher runs the same chunk-local pack from $j=0$ with the adapter disabled; the student writes the sink, each context chunk, and the query independently to h_{12} , then resumes the packed query through layers 12–35. If S_t is the teacher’s top-64 vocabulary set at query token t , then $p_t = \text{softmax}(z_{t,S_t}^T)$ and $q_t = \text{softmax}(z_{t,S_t}^S)$: both distributions are renormalized only over the same teacher-selected support, and all logits outside S_t are discarded by this approximate objective. We did not retain the teacher mass captured by S_t . The token-mean loss is

$$\mathcal{L}_{\text{distill}} = 0.6 \text{KL}(p||q) + 0.4 \text{KL}(q||p), \quad (2)$$

with no cross-entropy term. The rank-32 LoRA ($\alpha=64$, zero dropout) updates attention and MLP projections in layers 12–35; the Qwen3-8B backbone stays frozen. Training uses 4,000 bf16 DDP steps on 2,048-token PG-19 windows (Rae et al., 2019), without benchmark supervision, retrieval, or synthetic long-context data. Full optimizer, masks, reduction, and artifact details are in the appendix.

Selection and generation. The flagship selector is deterministic token-ID BM25 (Robertson and Zaragoza, 2009), chosen to avoid a separately trained retriever. It uses top-12, hop width four, and three frontier-expansion rounds: round 1 scores the question; later rounds concatenate tokens from newly selected chunks, score unselected chunks, and break ties by document index.

Frozen dense retrieval is reported only as a diagnostic. During generation, lower layers cache the query/generation prefix and upper layers cache the selected pack; the retrieved pack is not rerun for each output token. Exact masks, positions, and timing boundaries appear in the appendix.

5 Experiments

Setup. The principal model is frozen Qwen3-8B (Yang et al., 2025a), with $j=12$, 512-token chunks, iterative BM25 top-12, a BOS sink, and the rank-32 LoRA. Prompting is plain text without chat templates. We evaluate RULER (Hsieh et al., 2024), BABILong (Kuratov et al., 2024), LongEval (Li et al., 2023), six LongBench QA datasets (Bai et al., 2024), and all 1,986 LoCoMo questions (Maharana et al., 2024). Appendices A.2 and B give complete cells, prompts, sample counts, uncertainty, and integrity checks. We keep four timing boundaries separate: selected-pack model Read, store-ready online prefill, Write-inclusive pipeline, and external index/store cost.

5.1 Matched depth reuse

Table 2 is the central comparison. Raw-text replay ($j=0$) and CoMem receive the same selected chunks in the same order and use the same sink, mask, examples, and LoRA. Skipping the lower 12 layers reduces fixed-pack Read from 931.9 to 664.4 ms ($1.403\times$), but lowers the RULER macro from 99.19 to 96.07. CoMem is therefore not quality preserving. The continuous-prefix control exactly matches $j=0$, showing that upper-layer continuation can reproduce replay when supplied compatible states. Because it jointly restores document, cross-chunk, and query-side lower-layer interaction, the control is a fidelity ceiling rather than a single-factor explanation.

Natural-task scope checks are reported in full appendix tables rather than only in prose. The matched $j=0$ and $j=12$ rows score 97.2 versus 69.0 on LongEval, 41.59 versus 38.27 on LoCoMo, and 12.31 versus 12.15 on six LongBench QA datasets (Table 18); the task/length tables (Tables 20, 22, and 24) give the full $j=12$ and descriptive external-baseline breakdowns. Task-level BABILong results are mixed (Table 21). Because the PG-19 overlap audit is incomplete for several natural datasets, these are scope checks rather than contamination-free generalization or a heterogeneous leaderboard.

5.2 Amortization and equal-latency boundaries

The residual store costs 8,192 bytes/token and must first be written. In the write-once serving harness, CPU-pinned break-even is 8.9–10.9 queries at 32k for generated lengths up to 128 tokens; the full GPU/CPU grid is 5.5–10.9. The 512-token CPU case moves to 94 queries because decode dominates. At 128k and one generated token, break-even is 25.8/27.6 queries for GPU/CPU placement. Each value follows the displayed measured-time equation and includes fetch, model Read, and decode; for one-off or weakly reused documents, raw text is cheaper.

We then ask whether cheaper replay can buy enough extra evidence to improve quality at equal online latency. A latency-only calibration freezes maximum budgets of 10 replay chunks and 12 CoMem chunks; selectors may under-fill. With the same iterative-BM25 ranking, raw replay leads by 11.56 points. Replacing only the replay selector with frozen BGE changes the point difference to -1.00 , while the unchanged CoMem arm remains at 53.22. A hierarchical bootstrap that first resamples cells gives $[-18.67, -5.11]$ for BM25 and $[-10.67, 8.33]$ for BGE; leave-one-cell-out ranges are $[-13.13, -9.50]$ and $[-3.50, 1.75]$. Hence the decision-relevant conclusion is selector dependence: the BM25 disadvantage is stable, whereas the BGE aggregate is unresolved under task-cell dependence, and neither is a CoMem win. Appendix Table 8 exposes the cohort, weights, generation limits, split, absolute latencies, and bootstrap unit.

5.3 Bounded diagnosis

Holding retrieval, adapter, and upper-layer Read fixed, the 2×2 diagnostic separates lower-layer attention scope from Read positions. Document-contextual Write closes the gap under either position convention; document-origin positions alone make chunk-local states 4.5 points worse. The $+4.5$ interaction precludes an additive attribution, but bounds a major tested failure to the Write representation rather than position remapping.

Overlap-Write turns this control into a local deployable intervention: $w=32$ raises 92.5 to 98.5 and $w=128$ to 99.0 while preserving stored bytes and per-query work. Its measured wall-clock overhead exceeds the theoretical FLOP ratios, and it is evaluated only on this paired synthetic cohort. We

Operating point	Stored object	Replay	RULER↑	Read	Write	Store/token
Matched raw-text replay ($j=0$)	token IDs	36 layers	99.19	931.9 ms	on query	~4–8 B
CoMem + LoRA ($j=12$)	h_{12}	24 layers	96.07	664.4 ms	once	8,192 B
Continuous-prefix control	h_{12}	24 layers	99.19	—	non-deployable	8,192 B

Table 2: **The central result is a matched depth-reuse endpoint, not a cross-system ranking.** Quality uses the paired 15-cell RULER-B cohort ($n=1,500$); the $j=0$ – $j=12$ gap is 3.12 points with paired-bootstrap 95% CI [2.36, 3.93]. Read latency uses the same LoRA and a fixed selected pack (mean approximately 6.5k tokens; nominal cap 6,657) from 16k source contexts (three process medians); retrieval, persistent I/O, reusable Write, and decode are excluded. The continuous-prefix control supplies layer-12 states from a single causal lower-layer forward and exactly matches raw-text replay, showing that upper-layer continuation can be faithful with compatible states. The control jointly restores document, cross-chunk, and query-side lower-layer interaction, so it is a fidelity ceiling rather than a single-factor attribution.

Store	$G=1$	$G=32$	$G=128$	$G=512$
32k, CPU-pinned	8.9	9.2	10.9	94.0
32k, GPU-resident	8.4	7.7	5.5	—
128k, CPU-pinned	27.6	—	—	—
128k, GPU-resident	25.8	—	—	—

Table 3: **Measured repeated-query break-even Q^* .** G is generated tokens; a cell is the number of queries after which cumulative CoMem time is lower than matched $j=0$ raw-text replay. Both arms share selection, example, and pack; selection cancels in $Q^* = (W_{\text{CoMem}} - I_{j=0}) / (T_{j=0} - T_{\text{CoMem}})$. The 32k CPU cell uses $W_{\text{CoMem}}=2.253$ s and a 2.16 ms raw-token index build. Values derive from three process medians and include store fetch, model Read, and decode; raw component times and dispersion are archived. At 128k, only $G=1$ is retained, so no generation-length trend is claimed there. “—” denotes an unavailable or non-finite retained cell.

therefore claim a bounded diagnosis, not a universal repair.

Secondary scope checks. A frozen BGE-large-en-v1.5 diagnostic shows that retrieval recall can instead be the bottleneck (Appendix A.1). Store-ready dense-prefill and Write-inclusive measurements, synthetic 128k stress tests, and cross-scale/MoE ports are reported in the appendix. They support bounded-read operation and implementation portability, but do not alter the matched endpoint above.

Replay selector	$k_{\max}$	Replay / CoMem	Δ	Hier. 95% CI	LOCO range
Iterative BM25	10	64.78 / 53.22	−11.56	[−18.67, −5.11]	[−13.13, −9.50]
Frozen BGE	10	54.22 / 53.22	−1.00	[−10.67, 8.33]	[−3.50, 1.75]

Table 4: **Equal-latency quality is selector-dependent; CoMem does not win either aggregate.** The latency-only calibration freezes replay budget 10 against CoMem budget 12 within a predeclared $\pm 5\%$ band. Quality is the equal-weight mean of nine task-length cells ($n=100$ each), and Δ is CoMem minus replay. “Hier.” first resamples nine cells with replacement and then resamples the 100 paired examples within each selected cell (100,000 replicates, deterministic seeds). LOCO drops one cell and averages the other eight. The original pooled-IID intervals remain only as sensitivity: [−14.44, −8.67] (BM25) and [−4.67, 2.67] (BGE). The first 100 LoCoMo items are all conversation 0, so conversation-cluster resampling is not identifiable. Appendix Table 8 gives protocol, per-cell differences, and the fixed-cell stratified bootstrap.

Lower-layer Write scope	local Read pos.	document-origin Read pos.
Chunk local	92.5	88.0
Document contextual	100.0	100.0

Table 5: **Bounded context-scope $\times$ Read-position diagnosis.** Paired Qwen3-8B multikey macro over 8k/16k ($n=200$), with identical retrieval, LoRA, and upper-layer Read. Document context adds 7.5 points with local Read positions and 12.0 with document-origin positions; changing position alone lowers the chunk-local writer by 4.5 points. The +4.5 interaction means the factors are not additive. This attribution is limited to the displayed cohort.

Writer	Context	Macro $\uparrow$	Write FLOPs
Chunk local ($w=0$)	current chunk	92.5	$1.000\times$
Overlap ($w=32$)	32-token left prefix	98.5	$1.057\times$
Overlap ($w=64$)	64-token left prefix	98.5	$1.115\times$
Overlap ($w=128$)	128-token left prefix	99.0	$1.229\times$
Document-context control	full prefix	100.0	$O(N)$ forward

Table 6: **Lower-layer context is a focused repair, not a new headline result.** Paired Qwen3-8B results on RULER niah_multikey_1 at 8k and 16k ($n=200$ pooled). Every overlap width significantly exceeds the chunk-local writer; for $w=32$, the gain is $+6.0$ points, 95% CI $[3.0, 9.5]$. Persistent storage and Read/decode are identical across deployable rows. FLOPs are theoretical lower-12 Write ratios; measured wall-clock overhead is larger because of chunk-level execution. No natural-task overlap result is claimed.

6 Conclusion

CoMem makes transformer depth an explicit cross-query reuse axis: contexts are written to intermediate residuals, a bounded set is selected per query, and only the upper layers are resumed. The contribution is a matched internal operating-point measurement, not a claim of free compression or a leaderboard win. With identical evidence and adapter, $j=12$ yields $1.403\times$ faster model Read at a 3.12-point RULER cost, and its 8 KiB/token store pays off only after repeated queries. At equal online latency, iterative-BM25 raw replay is substantially more accurate, whereas the frozen-BGE difference is unresolved under cell-resampling uncertainty; CoMem wins neither aggregate. A paired context–position factorization further localizes one synthetic multikey gap to the lower-layer Write representation, and overlap recovers most of that local gap without enlarging the store or Read work. The next decisive step is a same-backbone, end-to-end comparison with PIC and learned modular-KV systems, plus natural-task validation of contextual writing. Until then, CoMem should be read as a transparent internal measurement of a depth-reuse design point with a bounded diagnosis.

Limitations

CoMem is evaluated primarily on one 8B backbone with a lexical selector and an English benchmark suite. Results on other Qwen scales and sparse MoEs support portability but are not matched replications, and several large-model cells use smaller sample counts. The flagship is one batch-8 training run. Two additional adapters

match its data budget but use effective batch 3, so the reported seed-plus-effective-batch check is not a clean estimate of training-seed variance.

The paper does not provide a same-backbone, same-hardware implementation of the closest PIC, chunk-KV repair, or learned modular-cache systems. Their published operating points use different models, tasks, training budgets, storage tiers, and latency boundaries, so Table 1 is taxonomic and our experiments establish an internal depth endpoint rather than competitive superiority. The fully matched causal evidence is a $j=0 \rightarrow 12$ two-point comparison; separately trained depths form a deployment curve, not one controlled multi-depth frontier.

The system assumes repeated queries over a relatively stable corpus. Its residual store grows linearly and is much larger than raw text or token IDs; for one-off queries, frequently edited documents, or low reuse, raw-text retrieval is usually preferable. Selector and external-store latency still grow with corpus size even though model-side Read is bounded. Reported crossover and speed measurements depend on hardware, kernels, batch size, storage tier, and the precise timing boundary. At 128k, the displayed break-even result covers only one generated token; longer-generation values were not retained and should not be inferred from the 32k grid.

Stored states are tied to the exact backbone, tokenizer, split depth, adapter, and lower-layer implementation; model upgrades generally require rewriting the store. Overlap-Write also weakens strict chunk independence because a local edit can invalidate the following chunk’s cached state. We do not evaluate residual quantization, eviction, or multi-tenant update contention.

Quality is task dependent. Bounded top- k retrieval imposes an evidence ceiling, lexical selection transfers poorly to low-overlap or non-whitespace languages, and residual readout can fail even when the evidence is present. Overlap-Write is tested on a focused multikey diagnostic rather than the natural benchmark suite, and its wall-clock Write overhead is not integrated into a repaired end-to-end frontier. It is not a universal quality fix. Results beyond the backbone’s native context window are positional-extrapolation stress tests.

The equal-latency mixture combines synthetic and natural diagnostics with equal cell weights. Under iterative BM25, raw replay wins under strat-

ified, hierarchical, and leave-one-cell-out analyses. With a frozen general dense retriever, the point estimate favors replay by one point but hierarchical and leave-one-cell-out analyses span both signs. The first 100 LoCoMo items all belong to conversation 0, so conversation-level clustering is impossible for this slice rather than silently treating it as 100 conversations. This selector and cell dependence precludes a selector-independent deployment conclusion.

LoCoMo’s primary semantic metric uses an undated mutable `gpt-4o` endpoint. We retain the evaluation date and saved item-level parsed decisions, and a 200-item DeepSeek-V3 audit preserves the ordering, but neither makes the full result exactly reproducible under a fixed judge snapshot. The PG-19 audit also found substantial overlap with InfiniteBench long-book support, prompting removal of that comparison. Equivalent overlap audits were not completed for all natural benchmarks, including NarrativeQA; natural-task results are scope checks, not contamination-free generalization estimates.

Our probes and controls characterize readout compatibility; they do not locate “understanding,” factual storage, or causal necessity at a particular layer. Long-form generation, code, multilingual and multimodal memory, dynamic updates/deletions, stronger learned retrievers, and production-scale multi-tenant scheduling remain open. The serving experiments report medians on single-query harnesses rather than concurrent p95 throughput or tail latency.

7 Ethical Considerations

CoMem changes the cost and storage organization of an underlying language model; it does not add a safety mechanism. It therefore inherits risks such as hallucination, biased generation, unsafe completion, and unintended disclosure of sensitive text selected from a memory store. Lower-cost access to long histories could also scale benign applications and deliberate misuse alike. Cached residual tensors may also expose source information through inversion or membership-inference attacks; they should receive protections at least as strong as the source text. We do not empirically evaluate these attacks, so the risk is a deployment concern rather than a demonstrated vulnerability. Their large persistent volume and possible cross-tenant reuse make authorization, isolation,

and deletion operational requirements rather than properties conferred by non-readable tensor storage. Deployments should encrypt and audit the persistent store, enforce access control, filter or redact sensitive sources before Write, support verifiable deletion, restrict retrieval to authorized material, and retain application-appropriate output monitoring and human review. Although bounded reads reduce accelerator memory and per-query computation, training and large-scale evaluation still consume energy; we report the measured hardware and final training budget in Appendix A.4.

We collect no new human-subject data and recruit no annotators. Our experiments use previously released model, corpus, and benchmark artifacts; LoCoMo’s released conversations are generated by paired language-model agents rather than newly collected private conversations. The anonymous release excludes model weights and benchmark text. Subject to benchmark and provider terms, it includes the adapter, evaluation scripts, prediction hashes or permissible shards, judge prompts/parsed decisions, and raw timing records; credentials and private API metadata are excluded.

References

- Shubham Agarwal, Sai Sundaresan, Subrata Mitra, Debabrata Mahapatra, Archit Gupta, Rounak Sharma, Nirmal Joshua Kapu, Tong Yu, and Shiv Saini. 2025. [Cache-Craft: Managing chunk-caches for efficient retrieval-augmented generation](#). *Proceedings of the ACM on Management of Data*.
- Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiansai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. 2024. [LongBench: A bilingual, multitask benchmark for long context understanding](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3119–3137.
- Zefan Cai, Yichi Zhang, Bofei Gao, Yuliang Liu, Tianyu Liu, Keming Lu, Wayne Xiong, Yue Dong, Baobao Chang, Junjie Hu, and Wen Xiao. 2025. [PyramidKV: Dynamic KV cache compression based on pyramidal information funneling](#). In *Conference on Language Modeling*.
- Chuangtao Chen, Grace Li Zhang, Xunzhao Yin, Cheng Zhuo, Bing Li, and Ulf Schlichtmann. 2026. [KV Packet: Recomputation-free context-independent KV caching for LLMs](#). *arXiv preprint arXiv:2604.13226*.
- Sabri Eyuboglu, Ryan Ehrlich, Simran Arora, Neel Guha, Dylan Zinsley, Emily Liu, Atri Rudra,

- James Y. Zou, Azalia Mirhoseini, and Christopher Ré. 2026. [Cartridges: Lightweight and general-purpose long context representations via self-study](#). In *International Conference on Learning Representations*, volume 2026, pages 42642–42687.
- Shiwei Gao, Youmin Chen, and Jiwu Shu. 2025. [Fast state restoration in LLM serving with HCache](#). In *Proceedings of the Twentieth European Conference on Computer Systems (EuroSys)*, pages 128–143.
- Aaron Grattafiori et al. 2024. [The Llama 3 herd of models](#). *arXiv preprint arXiv:2407.21783*.
- Momchil Hardalov, Gonzalo Iglesias, and Adrià de Gispert. 2026. [Cartridges at scale: Training modular KV caches over large document collections](#). *arXiv preprint arXiv:2606.04557*.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. 2024. [RULER: What’s the real context size of your long-context language models?](#) In *First Conference on Language Modeling*.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. [LoRA: Low-rank adaptation of large language models](#). In *International Conference on Learning Representations*.
- Junhao Hu, Wenrui Huang, Weidong Wang, Haoyi Wang, Tiancheng Hu, Qin Zhang, Hao Feng, Xusheng Chen, Yizhou Shan, and Tao Xie. 2025. [EPIC: Efficient position-independent caching for serving large language models](#). In *International Conference on Machine Learning*.
- Chao Jin, Zili Zhang, Xuanlin Jiang, Fangyue Liu, Shufan Liu, Xuanzhe Liu, and Xin Jin. 2025. [RAG-Cache: Efficient knowledge caching for retrieval-augmented generation](#). *ACM Transactions on Computer Systems*, 44:1–27.
- Yuri Kuratov, Aydar Bulatov, Petr Anokhin, Ivan Rodkin, Dmitry Sorokin, Artyom Sorokin, and Mikhail Burtsev. 2024. [BABILong: Testing the limits of LLMs with long context reasoning-in-a-haystack](#). In *Advances in Neural Information Processing Systems*, volume 37, pages 106519–106554.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuettler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33.
- Dacheng Li, Rulin Shao, Anze Xie, Ying Sheng, Lianmin Zheng, Joseph E. Gonzalez, Ion Stoica, Xuezhe Ma, and Hao Zhang. 2023. [How long can open-source LLMs truly promise on context length?](#) LM-SYS Org Blog.
- Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. 2024. [SnapKV: LLM knows what you are looking for before generation](#). In *Advances in Neural Information Processing Systems*, volume 37, pages 22947–22970.
- Manlai Liang, Mandi Liu, Jiangzhou Ji, Huaijun Li, Haobo Yang, Yaohan He, and Jinlong Li. 2025. [ILRe: Intermediate layer retrieval for context compression in causal language models](#). *arXiv preprint arXiv:2508.17892*.
- Shih-Ting Lin, Ashish Sabharwal, and Tushar Khot. 2021. [Readonce transformers: Reusable representations of text for transformers](#). In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*.
- Akide Liu, Jing Liu, Zizheng Pan, Yefei He, Gholamreza Haffari, and Bohan Zhuang. 2024. [MiniCache: KV cache compression in depth dimension for large language models](#). In *Advances in Neural Information Processing Systems*, volume 37, pages 139997–140031.
- Songshuo Lu, Hua Wang, Yutian Rong, Zhi Chen, and Yaohua Tang. 2025. [TurboRAG: Accelerating retrieval-augmented generation with precomputed KV caches for chunked text](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 6599–6612.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. 2024. [Evaluating very long-term conversational memory of LLM agents](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13851–13870.
- João Monteiro, Étienne Marcotte, Pierre-André Noël, Valentina Zantedeschi, David Vázquez, Nicolas Chapados, Christopher Pal, and Perouz Taslakian. 2024. [XC-Cache: Cross-attending to cached context for efficient LLM inference](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*.
- Kaleem Ullah Qasim, Jiashu Zhang, Muhammad Kafeel Shaheen, Razan Alharith, and Heying Zhang. 2026. The residual stream is all you need: On the redundancy of the KV cache in transformer inference. *arXiv preprint arXiv:2603.19664*.
- Jack W. Rae, Anna Potapenko, Siddhant M. Jayakumar, and Timothy P. Lillicrap. 2019. [Compressive transformers for long-range sequence modelling](#). *arXiv preprint arXiv:1911.05507*.
- Stephen Robertson and Hugo Zaragoza. 2009. [The probabilistic relevance framework: BM25 and beyond](#). *Foundations and Trends in Information Retrieval*, 4(1–2):1–174.

704	Jon Saad-Falcon, Amanpreet Singh, Luca Soldaini,	long-context extrapolation for LLMs with an effi-	760
705	Mike D’Arcy, Arman Cohan, and Doug Downey.	cient context memory. In <i>Advances in Neural In-</i>	761
706	2023. Embedding recycling for language models .	<i>formation Processing Systems</i> , volume 37, pages	762
707	In <i>Findings of the Association for Computational</i>	119638–119661.	763
708	<i>Linguistics: EACL 2023</i> .		
709	Zhenmei Shi, Yifei Ming, Xuan-Phi Nguyen, Yingyu	Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song	764
710	Liang, and Shafiq Joty. 2026. Discovering the gems	Han, and Mike Lewis. 2024b. Efficient streaming	765
711	in early layers: Accelerating long-context LLMs	language models with attention sinks . In <i>Interna-</i>	766
712	with 1000x input token reduction . In <i>Findings of</i>	<i>tional Conference on Learning Representations</i> .	767
713	<i>the Association for Computational Linguistics: ACL</i>	Hui Xie, Peng Xiao, Yutong Deng, Shuoran Dou, Jian	768
714	2026, pages 13839–13857.	Yang, and Jinyang Guo. 2026. SemPIC: Learning	769
715	Woomin Song, Sai Muralidhar Jayanthi, Srikanth Ro-	semantic position-independent KV caches . <i>arXiv</i>	770
716	nanki, Kanthashree Mysore Sathyendra, Jinwoo	<i>preprint arXiv:2607.28069</i> .	771
717	Shin, Aram Galstyan, Shubham Katiyar, and Sra-	Peng Xu, Wei Ping, Xianchao Wu, Lawrence	772
718	van Babu Bodapati. 2025. Compress, gather, and re-	McAfee, Chen Zhu, Zihan Liu, Sandeep Subrama-	773
719	compute: REFORMing long-context processing in	nian, Evelina Bakhturina, Mohammad Shoeybi, and	774
720	transformers . <i>arXiv preprint arXiv:2506.01215</i> .	Bryan Catanzaro. 2024. Retrieval meets long con-	775
721	Sijun Tan, Xiuyu Li, Shishir Patil, Ziyang Wu, Tian-	text large language models. In <i>International Confer-</i>	776
722	jun Zhang, Kurt Keutzer, Joseph E. Gonzalez, and	<i>ence on Learning Representations</i> .	777
723	Raluca Ada Popa. 2024. LLOCO: Learning long	An Yang et al. 2025a. Qwen3 technical report . <i>arXiv</i>	778
724	contexts offline . In <i>Proceedings of the 2024 Con-</i>	<i>preprint arXiv:2505.09388</i> .	779
725	<i>ference on Empirical Methods in Natural Language</i>	Xinyu Yang, Tianqi Chen, and Beidi Chen. 2025b.	780
726	<i>Processing</i> , pages 17605–17621.	APE: Faster and longer context-augmented genera-	781
727	Tencent Hunyuan Team. 2026. Hy3: An open	tion via adaptive parallel encoding . <i>arXiv preprint</i>	782
728	295b mixture-of-experts model with 21b acti-	<i>arXiv:2502.05431</i> .	783
729	vated parameters. https://huggingface.	Jiayi Yao, Hanchen Li, Yuhan Liu, Siddhant Ray, Yihua	784
730	co/tencent/Hy3 .	Cheng, Qizheng Zhang, Kuntai Du, Shan Lu, and	785
731	Jiahao Wang, Weiyu Xie, Mingxing Zhang, Boxing	Junchen Jiang. 2025. CacheBlend: Fast large lan-	786
732	Zhang, Jianwei Dong, Yuening Zhu, Chen Lin, Jinqi	guage model serving for RAG with cached knowl-	787
733	Tang, Yaochen Han, Zhiyuan Ai, Xianglin Chen,	edge fusion . In <i>Proceedings of the Twentieth Euro-</i>	788
734	Yongwei Wu, and Congfeng Jiang. 2026. From pre-	<i>pean Conference on Computer Systems</i> , pages 94–	789
735	fix cache to fusion RAG cache: Accelerating LLM	109.	790
736	inference in retrieval-augmented generation . <i>arXiv</i>	Zhenyu Zhang et al. 2023. H2O: Heavy-hitter oracle	791
737	<i>preprint arXiv:2601.12904</i> .	for efficient generative inference of large language	792
738	Qian Wang, Zahra Yousefijamarani, Morgan Lindsay	models. In <i>NeurIPS</i> .	793
739	Heisler, Rongzhi Gu, Bai Xiaolong, Shan Yizhou,		
740	Wei Zhang, Wang Lan, Ying Xiong, Yong Zhang,		
741	and Zhenan Fan. 2025. MEPIC: Memory effi-		
742	cient position independent caching for LLM serving .		
743	<i>arXiv preprint arXiv:2512.16822</i> .		
744	Weizhi Wang, Li Dong, Hao Cheng, Xiaodong Liu,		
745	Xifeng Yan, Jianfeng Gao, and Furu Wei. 2023.		
746	Augmenting language models with long-term mem-		
747	ory. In <i>Advances in Neural Information Processing</i>		
748	<i>Systems</i> .		
749	Yu Wang, Yifan Gao, Xiushi Chen, Haoming Jiang,		
750	Shiyang Li, Jingfeng Yang, Qingyu Yin, Zheng		
751	Li, Xian Li, Bing Yin, Jingbo Shang, and Ju-		
752	lian McAuley. 2024. MemoryLLM: Towards self-		
753	updatable large language models . In <i>Proceedings</i>		
754	<i>of the 41st International Conference on Machine</i>		
755	<i>Learning</i> , volume 235 of <i>Proceedings of Machine</i>		
756	<i>Learning Research</i> , pages 50453–50466. PMLR.		
757	Chaojun Xiao, Pengle Zhang, Xu Han, Guangxuan		
758	Xiao, Yankai Lin, Zhengyan Zhang, Zhiyuan Liu,		
759	and Maosong Sun. 2024a. InfLLM: Training-free		

Length	Dense prefill	CoMem prefill	Speedup	Dense peak	CoMem peak
8k	1.20 s	1.05 s	1.14×	18.8 GB	18.7 GB
32k	7.33 s	1.06 s	6.92×	25.0 GB	18.7 GB
128k	71.37 s	1.10 s	64.9×	50.0 GB	18.7 GB

Table 7: Select-first online prefill on one H20 (Qwen3-8B, bf16/SDPA, median of three after one warmup). Dense runs the stock full-context path; CoMem uses its rank-32 LoRA and a fixed 6,657-token top-12 pack. This cohort times online model prefill after the persistent memory is built; it excludes one-time document Write and external-store fetch. The speedup is therefore an online bounded-read operating point, not a production end-to-end or depth-only causal effect.

A Additional Results and Ablations

This appendix contains the detailed task tables, mechanism ablations, and cross-scale generality results summarized in the eight-page main paper. Appendix B separately documents aggregation, uncertainty, and the LoCoMo judge protocol.

Secondary dense-context operating point.

When the persistent store already exists, the select-first path reads a fixed 6,657-token pack at every source length. Table 7 reports 1.10 s and 18.7 GB at 128k versus 71.37 s and 50.0 GB for dense prefill on the same H20. The resulting 64.9×

Equal-latency audit. Table 8 reconstructs the complete saved protocol behind Table 4, including the nine cells, equal cell weights, disjoint latency calibration, generation limits, absolute TTFT, hardware, storage assumptions, all cell deltas, dependence-aware bootstraps, leave-one-cell-out sensitivity, and the original pooled-IID interval.

Content-depth probe protocol. For SST2, WiC, and RTE we extract every layer once from a fixed stratified pool (up to 3,000 train examples), then fit L2 logistic regressions on five independent 60/20/20 train/dev/test splits. Features are standardized; $C \in \{0.1, 1, 10\}$ is selected on development data. For layer accuracy $a(l)$, the knee is $\min\{l : a(l) \geq 0.98 \max_l a(l)\}$. The native readout instead applies the model’s own final norm and LM head to each layer using fixed verbalizers. Lexical uni/bi-grams, token length, random labels, majority class, and

balanced training serve as controls. On SST2, probe peak is about 0.90 versus 0.71 lexical, 0.56 position/majority, and 0.54 random-label accuracy; WiC/RTE are lower-selectivity and have wider per-task intervals. The reported confidence intervals are Student- t intervals over task-split knees, not training-seed uncertainty.

A.1 Mechanism ablations

Split depth and self-distillation. The probe values plotted in Figure 2 use five stratified data splits for Qwen and OLMo plus a matched-support Llama SST2 result; the protocol is specified above. Table 9 gives the frozen $j = 0/6/9/12$ quality sweep and same- j distilled control; Table 10 gives separately distilled $j = 6/9/12/18$ deployment points. Table 11 provides a separate same- j RULER control. Finally, Table 12 uses an HCache-style checkpoint path only as an interface-adaptation diagnostic, showing that the readout repair transfers beyond CoMem.

Selection. Table 13 compares oracle, BM25, recency, and reader attention. BM25 is near oracle on the lexical needle tasks and is more reliable than the tested training-free alternatives at long lengths. Variable tracking instead requires iterative expansion along literal variable links; a three-round CPU microbenchmark costs 188.8 ms at 128k, versus 41.0 ms for one-shot BM25, without an additional model forward. This is a low-overhead selector in the tested regime, not a constant-time one: lookup grows approximately linearly with store size (Table 25). A frozen BGE-large-en-v1.5 retriever is also evaluated on a focused cohort: it preserves strong hit-conditional reader accuracy but loses recall as context grows. We do not train or tune a retriever, so our results establish lexical and frozen-dense operating points rather than selector optimality.

Write-context factorization. The complete 2×2 control is reported in the main paper (Table 5). It confirms that the position-only intervention does not repair chunk-local states and that context and position interact.

Read interaction and chunk size. Table 16 compares full cross-chunk attention with block-diagonal reuse. Table 17 reports the chunk-size sweep; its entries are point estimates rather than a multi-seed stability ranking.

Protocol item	Setting	Auditable detail
Quality cohort	9 cells, $n=100/\text{cell}$	BABILong qa1/qa2 at 4k and 16k; LongEval at 8k and 16k; RULER niah_multikey_1 at 8k and 16k; first 100 LoCoMo items. Each cell contributes one equally weighted accuracy mean. The retained LoCoMo slice is entirely conversation 0, so its 100 questions cannot support a conversation-cluster bootstrap.
Generation/scoring	Greedy, no chat	BABILong: 20 tokens and official <code>compare_answers</code> ; LongEval: 16 tokens and numeric exact match; RULER: 48 tokens and official <code>string_match_all</code> ; LoCoMo diagnostic: 48 tokens and deterministic local substring/F1-threshold or abstention accuracy, not the GPT-4o headline judge.
Calibration split	Latency only, disjoint	Three RULER multikey 32k documents at reserved indices 900–902 (PYTHONHASHSEED=0, experiment seed 42), disjoint from quality indices 0–99. Integer $k \in \{2, 4, 6, 8, 10, 12, 14, 16, 20, 24\}$ is selected only by TTFT within $\pm 5\%$ of the CoMem budget-12 anchor. Here k is a maximum chunk budget; iterative BM25 may under-fill.
Timing	One NVIDIA H20; bf16/SDPA	Three independent processes per k , five warmups and 20 repetitions each; reported TTFT is the median of process medians. It includes online selection, raw/residual fetch, sink/query Write, and model prefill to first logits; decode, one-time document Write, and index construction are excluded. BM25 replay budget 10 is 703.7 ms versus CoMem 698.3 ms (GPU) / 699.8 ms (CPU); BGE replay is 697.8 ms versus 700.5 / 702.2 ms.
Cell deltas: BM25	CoMem(12) minus replay(10), points	qa1: -6 (4k), -17 (16k); qa2: -3 , -10 ; LongEval: -26 (8k), -28 (16k); RULER: -6 , -9 ; LoCoMo: $+1$. Aggregate -11.56 .
Cell deltas: BGE	CoMem(12) minus replay(10), points	qa1: -5 (4k), -21 (16k); qa2: 0 , -23 ; LongEval: 0 (8k), $+16$ (16k); RULER: $+1$, $+19$; LoCoMo: $+4$. Aggregate -1.00 .
Dependence-aware inference	100,000 paired bootstrap replicates	Stratified fixed-cell: keep all nine cells and resample exactly $n=100$ pairs inside each cell; intervals are $[-14.33, -8.78]$ (BM25) and $[-4.56, 2.56]$ (BGE). Hierarchical: resample nine cell labels, then resample $n=100$ pairs within every selected occurrence; intervals are $[-18.67, -5.11]$ and $[-10.67, 8.33]$. Leave-one-cell-out: $[-13.13, -9.50]$ (all 9 negative) and $[-3.50, 1.75]$ (7 negative, 2 positive). Deterministic base seed is 20260804.
IID sensitivity	Original v5 analysis	Pool all 900 paired differences and resample 900 pairs: $[-14.44, -8.67]$ (BM25) and $[-4.67, 2.67]$ (BGE). These intervals are retained for continuity but are not the task-mixture inference.

Table 8: **Protocol-complete equal-latency audit and dependence-aware reanalysis.** Phase B intentionally uses frozen BGE for raw replay and iterative BM25 for CoMem, so it tests selector sensitivity rather than a same-pack depth effect. The reanalysis reads only saved per-example scores; it runs no model. The anonymous release contains score-only exports (no benchmark text, gold answers, or predictions), the script, result JSON, and SHA-256 manifests.

Configuration	Split j	LoRA	LoCoMo Judge	BABILong mean	
				qa1 / qa2	qa5
Matched raw-text replay	0	no	41.59	66.3 / 38.9	63.3
CoMem frozen	6	no	32.78	44.6 / 22.6	53.9
CoMem frozen	9	no	29.15	42.4 / 19.6	55.6
CoMem frozen	12	no	24.52	33.4 / 18.0	60.3
CoMem distilled	12	yes	38.27	55.6 / 27.0	68.7

Table 9: Controlled comparison of retrieval, split depth, and self-distillation using Qwen3-8B. All configurations follow the same plain-text prompting protocol, with model-specific chat templating disabled (`chat_template=False`), and retrieve the same top-12 pack. For the frozen configurations, readout fidelity decreases as the split moves deeper. At the same split $j=12$, the rank-32 LoRA improves the LoCoMo Judge score by 13.75 points and the BABILong qa1, qa2, and qa5 means by 22.2, 9.0, and 8.4 points, respectively. BABILong entries are rounded task means from the underlying 21 task-length cells; the raw-cell $j=0$ macro is 56.14. The $j=0$ configuration performs full-depth recomputation over the retrieved pack, providing a selected-pack reference without lower-layer reuse and thus no depth-based reduction in query-time computation.

Split j	LoRA params	RULER B	Full — deployed (95% CI)	LoCoMo	Read (ms)	Read speedup	Write (ms)
6	72.745M	98.29	0.91 [0.49, 1.39]	40.38	830.3	1.166×	133.4
9	65.470M	97.55	1.65 [1.04, 2.31]	39.02	748.3	1.273×	196.9
12	58.196M	96.07	3.12 [2.36, 3.93]	38.27	664.4	1.403×	—
18	43.647M	55.41	43.79 [41.77, 45.85]	28.65	499.5	1.807×	390.3

Table 10: Multi-depth deployment curve. Each split has a separately trained rank-32 adapter on layers $[j:36]$ under the same data order, seed, steps, optimizer, and KL recipe. RULER uses Cohort B ($n=1,500$ paired examples); gap CIs are paired bootstrap intervals against full replay with that depth’s adapter, and all McNemar tests reject equality ($p \leq 3.05 \times 10^{-5}$). LoCoMo reports the full-set GPT-4o judge point estimate ($n=1,986$). Read/Write use a fixed 16k same-pack input and three process medians; retrieval, I/O, and decode are excluded, and total latency remains decode-dominated. The matched fixed-pack Write value was not retained for the earlier $j=12$ artifact. Adapter spans and parameter counts differ, so this is not a training-compute-matched depth ablation.

Task	Adapter	8k	16k	32k
niah_single_2	off	36	7	22
	on	100	100	100
niah_multikey_1	off	8	3	1
	on	91	95	92

Table 11: Separate controlled RULER self-distillation ablation on Qwen3-8B at the same split $j=12$ ($n=100$ per cell). Each on/off pair uses the same task, length, selector, and cohort, so the difference isolates the adapter. Table 9 provides the broader same- j LoCoMo/BABILong control used in the main argument.

Task	Len	BM25	Recency	ReaderAttn	Oracle
niah_single	8k	100	100	100	100
	16k	100	100	100	100
	32k	100	82	73	100
niah_multikey	8k	99	98	97	100
	16k	99	88	90	100
	32k	99	54	60	100
var-track	8k	99.4	99.2	99.8	N/A
	16k	92.6	92.4	92.4	N/A
	32k	32.0	41.2	27.8	N/A

Table 13: Single-pass selector sweep on RULER ($n=100$) without a chat template; each entry is the peak over $k \in \{4, 8, 12, 16, 24\}$. Oracle retrieval keeps both needle tasks at 100, showing that their long-range gap is selection rather than readout loss. The variable-tracking rows belong to this specific selector-sweep configuration and should not be conflated with the focused top-12 flagship in Table 15. Oracle is undefined for a multi-chunk reference chain and is therefore not reported there.

Configuration	LoRA	LoCoMo Judge
HCache-style ($j=12$)	no	13.29
HCache-style ($j=12$)	yes	31.17

Table 12: Interface-adaptation diagnostic on an HCache-style, retrieval-free checkpoint path. The two configurations share the $j=12$ split and evaluation harness, isolating the adapter. The gain shows that aligning intermediate states to native upper-layer readout transfers beyond CoMem; this is not a head-to-head long-term-memory comparison.

Task / length	qa1 16k	qa2 16k	LongEval 16k	RULER multikey 16k	LoCoMo
Dense recall@12	.52	.45	.58	.81	.70
Raw-text reader quality	.45	.31	.57	.82	.18

Table 14: **Frozen dense-retriever diagnostic.** BGE-large-en-v1.5 uses CLS pooling, L2 normalization, cosine search, the same 512-token chunks, and a Qwen3-8B $j=0$ reader without LoRA ($n=100$ per cell). Long-context recall can be the bottleneck even when hit-conditional reading remains strong; this is a fixed top-12 diagnostic, not a trained-retriever comparison. The separate equal-latency selector audit is Table 4.

Method	RULER task	8k	16k	32k	64k	128k
CoMem	niah_single_2	100.0	100.0	99.0	99.0	100.0
	var-track	96.6	97.6	98.8	99.0	95.8
StreamingLLM	niah_single	85.0	36.0	20.0	10.0	2.0
	var-track	41.8	2.2	0.6	0.2	0.8

Table 15: Equal-budget, no-chat RULER comparison ($n=100/\text{cell}$). CoMem and StreamingLLM use the same nominal budget and scorer but not paired examples; CoMem retrieves about 6.5k relevant tokens, whereas StreamingLLM keeps recent tokens.

Task	Full attn.	Block-diag.	Δ
RULER niah_single	100 / 100	100 / 96	0 / +4
RULER niah_multikey	96 / 94	60 / 32	+36 / +62
BABILong qa2	36 / 20	17 / 13	+19 / +7
BABILong qa5	78 / 69	78 / 55	0 / +14

Table 16: Full cross-chunk attention versus block-diagonal reuse (8k/16k, no chat). RULER uses $n=50$ and BABILong $n=100$ with identical top-12 retrieval. Joint recomputation matters most for multi-fact tasks.

chunk	read tokens	uncached step (s/token)	8k	16k	32k	64k
128	1,665	0.19	91.0	90.0	81.0	85.0
256	3,329	0.37	80.0	90.0	90.0	94.0
512	6,657	0.72	89.0	95.0	97.0	94.0
1024	13,313	1.60	100.0	89.0	92.0	94.0

Table 17: Chunk-size trade-off on RULER niah_multikey ($n=100$, no chat). Packed-step cost grows approximately linearly with read length; accuracy values are point estimates, not a multi-seed claim.

A.2 Full chat-template-free benchmark comparisons

Tables 19–22 give the full audited comparisons for RULER, LongEval, BABILong, and LongBench. Main rows use the unified no-chat protocol; they include KV-Direct (Qasim et al., 2026), In-fLLM (Xiao et al., 2024a), StreamingLLM (Xiao et al., 2024b), and MemoryLLM (Wang et al., 2024). MemoryLLM uses its released Llama-3 checkpoint (Grattafiori et al., 2024), while the marked LLoCO row uses its released native setup (Tan et al., 2024). These are descriptive external references rather than matched rankings. The all-method LoCoMo comparison is Table 24; judge details are in Appendix B.

Method	RULER [†] 15-cell	LongEval [†] 8k–128k	LongBench [†] 6-QA F1	BABILong [†] qa1/2/5 macro	LoCoMo Judge [†] full set
Matched raw-text replay ($j=0$)	99.20 ^B	97.2	12.31	56.14	41.59
CoMem + LoRA ($j=12$)	97.05 ^A / 96.07 ^B	69.0	12.15	50.43	38.27
<i>Unmatched descriptive references; do not compare columns as a ranking</i>					
KV-Direct	78.80 ^A	65.2	12.17	63.00	34.59
InfLLM	77.83 ^A	21.6	11.86	59.52	22.21
StreamingLLM	23.37 ^A	30.8	11.11	48.14	25.63
MemoryLLM [†]	16.55 ^A	13.6	9.01	30.00	16.11

Table 18: **Descriptive cross-benchmark scope table, not a method leaderboard.** Only the $j=0$ and $j=12$ rows form the paper’s matched depth comparison. Other rows differ in interface, context treatment, and sometimes backbone; [†]MemoryLLM uses a released 7B model. RULER superscripts are distinct cohorts and are never pooled or ranked across superscripts.

Method	Task	8k	16k	32k	64k	128k
CoMem + LoRA	single-2	100	100	99	99	100
	multikey	95	94	97	91	93
	var-track	96.6	97.6	98.8	99.0	95.8
CoMem frozen ($j=9$)	single	96	99	99	99	96
	multikey	44	44	59	30	48
	var-track	45	38.8	36.4	31.2	25.8
KV-Direct	single	100	100	100	100	0
	multikey	100	100	98	88	0
	var-track	100	100	99.8	96.2	0
InfLLM	single	100	100	92	61	57
	multikey	100	79	65	45	20
	var-track	100	96.8	90.6	81.8	79.2
StreamingLLM	single	85	36	20	10	2
	multikey	83	34	18	13	4
	var-track	41.8	2.2	0.6	0.2	0.8
<i>External persistent-memory reference (different backbone)</i>						
MemoryLLM [†]	single	29	40	37	21	21
	multikey	28	24	25	12	10
	var-track	0	1.2	0	0	0

Table 19: Full chat-template-free RULER comparison using official `string_match_all` ($n=100$ per cell). KV-Direct places the full context in the read with native RoPE and no YaRN; its 64k/128k cells are unextended stress references rather than a length-extended upper bound. CoMem keeps a bounded retrieved pack. This is **Cohort A**: every method uses `niah_single_2`, `niah_multikey_1`, and `variable_tracking`, with $n=100$ and the official scorer. Its CoMem macro is 97.05. Cohort B instead uses `niah_single_3` to match the YaRN arm (Table 23) and yields 96.07; the cohorts are never pooled in a direct comparison. [†]MemoryLLM is the principal external persistent-memory reference and uses its released Llama-3-8B-chat checkpoint; its different backbone precludes matched attribution.

Method	8k	16k	32k	64k	128k	Mean
CoMem + LoRA	69	75	64	67	70	69.0
CoMem frozen ($j=9$)	8	0	4	0	4	3.2
KV-Direct	100	96	92	38	0	65.2
StreamingLLM	86	34	18	10	6	30.8
InfLLM	60	30	12	4	2	21.6
MemoryLLM [†]	22	22	16	6	2	13.6

Table 20: LongEval register-content accuracy. Unmarked rows use the unified chat-template-free protocol. Means use the common 8k–128k support. KV-Direct uses native RoPE without YaRN, so its 64k/128k cells are unextended stress references and the five-length mean is not a length-extended upper bound. CoMem’s separate six-length headline is 72.83 after including its 4k score of 92. The frozen CoMem control used 48 generation tokens rather than 16; both budgets exceed the single-number answer length. [†]MemoryLLM is the external persistent-memory reference and uses a different chat-tuned backbone, so its row is not a matched comparison.

Method	Task	0k	1k	2k	4k	8k	16k	32k	Mean
CoMem + LoRA	qa1	98	80	68	68	46	17	12	55.6
	qa2	26	44	43	44	23	8	1	27.0
	qa5	68	76	76	75	68	60	58	68.7
CoMem frozen ($j=9$)	qa1	98	59	71	57	7	1	4	42.4
	qa2	53	17	28	26	8	4	1	19.6
	qa5	70	74	65	60	41	39	40	55.6
KV-Direct	qa1	98	84	80	74	80	72	63	78.7
	qa2	58	53	50	46	49	49	37	48.9
	qa5	71	73	62	59	65	42	58	61.4
InfLLM	qa1	97	84	81	72	69	52	34	69.9
	qa2	51	58	41	47	41	39	30	43.9
	qa5	69	68	63	66	73	60	55	64.9
StreamingLLM	qa1	97	83	80	71	23	27	12	56.1
	qa2	49	56	44	49	19	11	4	33.1
	qa5	68	65	68	65	39	47	34	55.1
MemoryLLM [†]	qa1	52	46	35	28	23	17	12	30.4
	qa2	37	29	17	18	21	17	11	21.4
	qa5	50	45	39	35	35	34	29	38.1

Table 21: BABILong accuracy ($n=100$ per task-length cell), evaluated with the official `compare_answers` scorer and `TASK_LABELS`. Within the native context window, full-context KV-Direct is substantially stronger on qa1 and qa2, exposing the compression tax, whereas distilled CoMem achieves the highest qa5 mean, ahead of InfLLM (68.7 vs. 64.9). All methods are evaluated under the unified plain-text prompting protocol, with model-specific chat templating disabled (`chat_template=False`). [†]MemoryLLM is the principal external persistent-memory reference and uses its released Llama-3-8B-chat checkpoint. Its different backbone precludes matched attribution. We retain the uniform no-chat row and make no prompt-sensitivity claim without a paper-retained matched table.

Method	NQA	Qasper	Hotpot	2Wiki	MultiF	Musique	Macro
CoMem + LoRA	4.12	11.01	11.62	12.83	25.41	7.91	12.15
CoMem frozen ($j=9$)	4.63	11.23	9.41	10.71	22.05	5.72	10.63
KV-Direct	3.70	11.82	12.68	12.03	25.30	7.49	12.17
InfLLM	2.99	11.35	12.08	12.50	25.33	6.94	11.86
StreamingLLM	3.49	11.64	11.02	12.19	22.42	5.88	11.11
MemoryLLM [†]	3.13	8.46	8.76	10.27	17.43	5.98	9.01
LLoCO [‡]	24.21	24.45	44.24	—	—	—	30.97

Table 22: Official LongBench QA token F1 by dataset. Except for the explicitly marked LLoCO diagnostic, methods use the same plain-text no-chat protocol. CoMem with LoRA scores 12.15 and full-context KV-Direct scores 12.17; because only CoMem uses the split-interface adapter, this near-equality is descriptive rather than a training-budget-matched compression estimate. The frozen CoMem configuration reports adapter-free performance at $j=9$. NQA denotes NarrativeQA, and MultiF. denotes MultiFieldQA-en. [†]MemoryLLM is the principal external persistent-memory reference and uses its released Llama-3-8B-chat checkpoint. Its different backbone precludes matched attribution. We retain the uniform no-chat row and make no prompt-sensitivity claim without a paper-retained matched table. [‡]LLoCO uses LLaMA-2-7B AutoCompressor, released per-domain supervised LoRAs, fp16, and its native task prompts. Released weights cover only NarrativeQA/Qasper/HotpotQA on our 200-example sets; 30.97 is their three-task macro and is not comparable to the six-task macro column.

Task	Method	8k	16k	32k	64k	128k
single needle	KV-Direct	100	100	100	100	0
	KV-Direct+YaRN [†]	100	100	100	100	100
	CoMem+LoRA	100	91	97	98	98
	CoMem+LoRA+YaRN [†]	100	90	97	97	98
multikey	KV-Direct	100	100	98	88	0
	KV-Direct+YaRN [†]	98	94	96	91	89
	CoMem+LoRA	94	91	99	90	93
	CoMem+LoRA+YaRN [†]	82	85	94	88	93
variable tracking	KV-Direct	100	100	99.8	96.2	0
	KV-Direct+YaRN [†]	99.2	99.4	26.6	67.2	57.8
	CoMem+LoRA	96.2	98.0	98.2	98.6	99.0
	CoMem+LoRA+YaRN [†]	88.6	90.0	89.4	95.0	93.6

Table 23: Matched- $n=100$ Qwen3-8B RULER scaling. [†]YaRN factor 4 gives an effective 163,840-token window. At 128k, unextended-backbone CoMem+LoRA scores 98/93/99.0 on single-needle, multikey, and variable tracking, versus 100/89/57.8 for KV-Direct+YaRN. Applying YaRN to CoMem changes these to 98/93/93.6, showing an in-window position-scaling tax on variable tracking.

Method	Judge	Judge _{1:4}	F1	Substr. acc.
CoMem + LoRA ($j=12$)	38.27	48.64	9.15	23.36
KV-Direct	34.59	43.83	9.02	22.36
StreamingLLM	25.63	31.04	7.67	13.75
InfLLM	22.21	26.43	7.39	13.34
MemoryLLM [†]	16.11	20.65	5.91	9.52

Table 24: LoCoMo comparison on all 1,986 chat-template-free examples. Judge_{1:4} is GPT-4o accuracy on the 1,540 answerable items; the full Judge column folds in local abstention scoring for 446 adversarial items. F1 and substring accuracy are deterministic lexical diagnostics; semantic Judge is the primary metric. All rows use the same scoring protocol. The same- j adapter effect is isolated in Table 9. On paired answerable items, distilled CoMem exceeds KV-Direct by 4.81 points (item-bootstrap 95% CI [2.34, 7.34]); a 10-conversation cluster bootstrap gives [1.27, 8.32] and 8/10 observed conversation differences favor CoMem. An independent DeepSeek-V3 audit preserves the ordering. [†]MemoryLLM is the external persistent-memory reference and uses its official chat-tuned Llama-3 checkpoint; its different backbone precludes matched attribution. The paper retains only its uniform no-chat evaluation and makes no native-chat sensitivity claim.

A.3 Store scaling and failure boundaries

Table 25 tests the “bounded read over extensible store” claim directly. Retrieval is invariant when the required evidence fits top-12, but lookup and index construction still scale with store size. Beyond the budget, recall is physically capped; co-keyed multivalued evidence falls below even this cap (recall 0.62 at $E=8$), and a common-word-frequency task scores zero as coverage falls from 4.5% at 128k to 1.2% at 512k. These negative re-

sults delimit the claim: bounded retrieval is not global aggregation, and successful retrieval does not guarantee multi-hop reasoning.

We audited the PG-19 adapter corpus against the InfiniteBench long-book support using normalized 13-gram containment sketches. The conservative 0.8 threshold flags 24/86 books (163/580 records), but containment is sharply bimodal: 31 books are below 0.10, no book lies in 0.18–0.60, and 54 are at least 0.60. Any cut in 0.20–0.60 therefore flags 54/86 books and 387/580 records; anonymized character names make the 0.8 count an undercount. Existing predictions were unavailable for a strict-clean-subset rescore (113 QA and 76 choice records), so we remove the book-quality comparison rather than treat it as out-of-domain evidence. This does not affect the synthetic store-scaling results above. We did not complete equivalent overlap audits for every natural benchmark, including NarrativeQA; those results should therefore be interpreted as benchmark performance rather than contamination-free generalization estimates.

A.4 Reproducibility details

Tables 26–28 record the configuration, training, and evaluation settings used by the Qwen3-8B flagship. The archived configuration files and evaluation scripts accompany the code release.

Artifacts, licenses, and data scope. The study evaluates English text only. It uses the frozen released Qwen/Qwen3-8B checkpoint (Apache-2.0); we do no backbone continued pretraining. It also uses PG-19 (Rae et al., 2019),

Store	NIAH recall	NIAH score	VT recall	VT score	Read tokens	BM25 (ms)
128k	1.00	100	1.00	18	6,211 / 6,463	80 / 20
256k	1.00	100	1.00	34	6,209 / 6,463	164 / 39
512k	1.00	100	1.00	40	6,209 / 6,464	328 / 81
1M	1.00	100	1.00	54	6,209 / 6,463	697 / 170
2M	1.00	90	1.00	38	6,211 / 6,463	1,407 / 357
4M	1.00	100	1.00	34	6,208 / 6,463	2,852 / 725

Table 25: Store scaling with a fixed top-12 read. Values separated by “/” are single-evidence NIAH / five-link variable tracking (VT). Evidence recall ($n=20$) stays perfect and the model read stays about 6.2–6.5k tokens from 128k to 4M stored tokens; BM25 lookup grows approximately linearly. Generation scores use $n=10$ and reveal that VT reasoning remains noisy even when all five evidence chunks are retrieved. At 128k, increasing required VT evidence from 12 to 16/24/32 gives recall 1.0 to 0.75/0.50/0.375, exactly $\min(1, 12/E)$.

Layer	Setting	Value
Backbone	Architecture	Frozen Qwen/Qwen3-8B; revision b968826d9c46 (full hash in artifact manifest); $L=36$, $d=4096$, 32 query heads, 8 KV heads, bf16, SDPA.
Backbone	Positions	Native window 40,960; RoPE $\theta=10^6$; no active RoPE scaling or YaRN.
Write	Cached state	Split $j=12$; disjoint 512-token chunks; chunk-local positions restart at zero; one bf16 residual tensor h_j per token.
Read	Pack	[BOS; selected h_j ; $h_j(q)$] in document order, with fresh contiguous RoPE and full causal cross-chunk attention. The nominal cap is $1 + 12 \times 512 + 512 = 6,657$ tokens; actual RULER means are about 6.2–6.5k because tail chunks and queries can be shorter.
Retrieval	BM25	Token-ID documents and bare-question token IDs; $k_1=1.5$, $b=0.75$, Robertson IDF with +1 smoothing; no lowercasing, stemming, or stop-word removal.
Retrieval	Iteration	$\text{rounds}=0$ means automatic $\lceil k/h \rceil$ rounds, not one-shot. Each round uses newly selected chunks as the next frontier and deduplicates the selected set.
Retrieval	Width	All five benchmarks fix $k=12$, hop width $h=4$, and therefore three automatic rounds. Chunk size 512 and the BOS sink are also fixed.
Read	Mask	The pack is one causal sequence: the final query sees all chunks; later chunks may attend to earlier chunks. In the block-diagonal ablation, chunks see only themselves and the sink, while the final query still sees all chunks.
Generation	Decoding	Greedy decoding (no sampling, temperature, top- p , or beams); no chat template and no thinking mode.
Adapter-free arm	Deployment	Frozen backbone, no LoRA, $j=9$; all other memory and evaluation settings are held fixed.

Table 26: Flagship CoMem inference configuration. The retrieval and read configuration is frozen across all five benchmarks; only benchmark harness parameters differ.

RULER (Hsieh et al., 2024), BABILong (Kuratov et al., 2024), LongBench (Bai et al., 2024), LongEval/LongChat (Li et al., 2023), and LoCoMo (Maharana et al., 2024). RULER and LongChat code are Apache-2.0; the LongBench repository is MIT-licensed, while its component datasets retain their upstream terms; BABILong code is Apache-2.0 and incorporates BSD-licensed bAbI tasks; LoCoMo is CC BY-NC 4.0. PG-19 contains pre-1919 Project Gutenberg books, for which the dataset page does not assert one uniform corpus license, so we do not redistribute its text. We likewise redistribute no model weights, benchmark examples, or private API responses. Subject to upstream terms, the anonymous archive includes the adapter, source, documentation, pinned requirements, prediction hashes or permissible shards, judge prompts/parsed outputs, raw timing records, and third-party notices. Users obtain all external artifacts from their original providers and remain responsible for their terms. Table 28 reports the evaluated supports and sample counts; LoCoMo category denominators

are in Appendix B.

Software environment. The archived environment specifies Python 3.10+, PyTorch 2.10, Transformers 5.5.4, PEFT 0.10+, Datasets 2.14+, bf16, and SDPA. Evaluation uses the benchmark-specific official scorers named in Table 28; the archive records all non-default flags and provides a CPU correctness gate.

Compute accounting. The final adapter run used one node with 8 H20 GPUs for about 22 minutes, or approximately 2.9 H20 GPU-hours. Table 27 and this section report model size, trainable parameters, hardware, and measured timing protocol. Total GPU-hours across preliminary probes, failed runs, baseline generation, and all ablations were not consistently logged; we report this omission rather than extrapolating an unreliable total.

Efficiency measurement. The Write-inclusive cohort uses one NVIDIA L20A with PyTorch 2.10, Transformers 5.5.4, bf16, SDPA, and the rank-32 LoRA enabled, reporting medians of three repeti-

Setting	Value
Adapter	Rank 32, $\alpha=64$, dropout 0; Q/K/V/O and gate/up/down projections in layers 12–35 only; 58.20M trainable parameters (0.71%). Final weight SHA-256: dd09cd17457c63578c0f38dab79b287ab5da6e3f14c119aedaefec1c34400536f.
Teacher and objective	The same Qwen3-8B with adapters disabled and $j=0$ serves as the frozen teacher. At every query token, the teacher top-64 vocabulary indices define a shared support; teacher and gathered student logits are each softmax-renormalized on that support, and outside-support logits are discarded. $\mathcal{L} = 0.6 \text{KL}(p\ q) + 0.4 \text{KL}(q\ p)$ at temperature 1, averaged over query tokens, with no CE term. Retained teacher top-64 probability mass was not logged.
Data	Plain PG19 train text; no synthetic long-context examples, retrieval, task labels, or benchmark supervision. Training uses 2,048-token windows, formed as four 512-token chunks.
Forward construction	Each 2,048-token window is split into three context chunks and one query chunk. Teacher and student use the same BOS sink, chunk token IDs, order, and query segment. The teacher replays every chunk from layer 0 with the adapter disabled; the student writes chunks independently through layers 0–11 without gradient and resumes one causal packed sequence through layers 12–35 with the LoRA. Loss is computed only on all query tokens.
Optimization	AdamW, $\beta = (0.9, 0.95)$, weight decay 0, peak LR 8×10^{-5} , 100-step linear warmup, cosine decay to zero, gradient clipping 1.0, seed 42, bf16, no gradient checkpointing.
Schedule	4,000 steps on 8 GPUs, one sample per GPU and no gradient accumulation (global batch 8), totaling about 65.5M tokens.
Compute	One node with 8×NVIDIA H20 GPUs. Logged throughput is about 24.5 samples/s, implying roughly 22 minutes wall-clock; training peak memory was not recorded.

Table 27: Self-distillation and LoRA training configuration. The adapter name’s “4k” denotes 4,000 optimization steps; the training sequence length is 2,048 tokens.

Benchmark	Tasks and support	Generation	Scoring and aggregation
RULER	Single needle, multikey, and variable tracking; 8k–128k; 100 examples/cell; 8 shards; seed 42.	48 tokens; 60 for variable tracking.	Official case-insensitive <code>string_match_all</code> ; equal-weight macro over 15 cells.
LongEval	Lines retrieval; 4k–128k; 100 examples/length; 8 shards; seed 1234.	16 tokens for the flag-ship; 48 for frozen controls.	First numeric answer exact match; six-length macro (five-length 8k–128k macro for matched baseline tables).
LongBench	Six QA datasets; 1,150 examples total; 8 shards; seed 42.	32/64/128 tokens by dataset.	Official multi-reference token F1; equal-weight macro over six datasets.
BABILong	qa1/qa2/qa5; 0k–32k; 100 examples/cell; 4 shards; seed 42.	20 tokens.	Official <code>compare_answers</code> with <code>TASK_LABELS</code> .
LoCoMo	All 1,986 QA pairs from 10 conversations; 4 shards.	48 tokens for the main 8B comparison; 512 for the cross-scale audit.	GPT-4o judge on categories 1–4; official local abstention rule for category 5; full-set judge is the headline.

Table 28: Benchmark-level evaluation settings. Every main comparison uses `chat_template=False`. MemoryLLM is the external persistent-memory reference, evaluated with its released chat-tuned backbone.

tions. Full-context prefill times the complete forward; CoMem times document Write, CPU retrieval, query Write, and upper-layer Read through logits. The timer excludes one-time BM25 index construction and external-store I/O; Table 33 measures the latter separately. At 128k, full-context prefill is 6.035 s and peaks at 89.39 GB, while CoMem takes 2.202 s and peaks at 18.54 GB. Cached decode uses 20 greedy steps.

An earlier representative operating-point cohort is a separate one-H20 bf16/SDPA sweep; its A/B quality values are not a single matched Pareto frontier. Retrieved rows share top-12 IDs, a 6,657-token nominal pack (measured ~ 6.5 k), and chunk size 512. Write-all timing processes the complete store once; Read is per query. The $j=0$ text-retrieval path has no adapter, whereas $j=12$ uses the flagship LoRA; hence the 0.33 s Read difference is an operating-point comparison rather than a depth-only effect. Its 5.74 s extra Write implies a 17–20-query analytic break-even in that earlier component cohort. The main serving experiment instead measures write-once/fetch/reuse

directly. At 32k, CPU-pinned break-even is 8.9–10.9 queries for generations up to 128 tokens, while the complete GPU/CPU grid is 5.5–10.9; at 128k the retained one-token points are 25.8–27.6. The difference from the earlier component cohort comes from source length, generation length, storage tier, and timing boundary. CoMem stores 8 KiB/token (about 1 GiB at 128k), far more than text but $18\times$ less than full KV. Main timing cohorts keep the store in HBM; Table 33 separately measures placement beyond HBM, so its I/O microbenchmark is not added to model latency.

Retained-KV references. Table 30 reports SnapKV and PyramidKV (Li et al., 2024; Cai et al., 2025) under their full-prefill boundary. They answer a different systems question from cross-query persistence and are not treated as nearest modular-cache baselines.

Paired quality-latency replay control. Table 31 uses a separate H20 harness. The two arms share the Qwen3-8B checkpoint, flagship LoRA SHA-256 and all 168 mounted modules, example IDs, decode parameters, and per-example re-

Suite	Adapter	Seed 42	Seed 1	Seed 2	Mean	SD
RULER 9-cell macro	batch 8 / 3 / 3	98.80	99.42	98.76	98.99	0.37
BABILong 9-cell macro	batch 8 / 3 / 3	38.44	40.33	41.44	40.07	1.52
RULER multikey 3-length mean	batch 8 / 3 / 3	96.67	98.67	96.67	97.33	1.15
BABILong qa5 3-length mean	batch 8 / 3 / 3	66.00	69.33	71.00	68.78	2.55

Table 29: **Seed-plus-effective-batch robustness, not clean seed variance.** All adapters use the same Qwen3-8B split, PG-19 data budget, and rank-32 objective; seed 42 is the flagship effective-batch-8 run, while seeds 1 and 2 use effective batch 3 with matched examples seen. Evaluation uses identical cells and supports: RULER single/multikey/variable-tracking at 32k/64k/128k ($n=50/\text{cell}$) and BABILong qa1/qa2/qa5 at 4k/16k/32k ($n=100/\text{cell}$). SD is the sample SD across the three run-level cell macros. These reduced-support cells are not the 15-cell RULER-B headline, and no LoCoMo or LongEval multi-run aggregate was retained.

(a) Quality at a 6,657-token/KV budget						(b) Full-prefill systems cost on one H20				
Method	Native A macro	Full-prompt 64k/128k			LoCoMo F1/acc	Method	Len.	ms	GB	ms/tok
		single	multikey	tracking						
CoMem+LoRA	97.05	99/100	91/93	99.0/95.8	9.15/23.36	SnapKV	8k	1,213	18.6	24.7
SnapKV	72.33 [†]	100/100	94/91	80.0/84.8	9.21/22.05		32k	6,342	25.7	24.6
PyramidKV	72.32 [†]	100/100	93/89	88.6/86.8	9.13/22.10	PyramidKV	128k	51,520	54.3	24.7
							8k	1,203	18.6	24.7
							32k	6,301	25.7	26.7
							128k	51,520	54.3	26.8

Table 30: Standard selected/compressed-KV baselines at CoMem’s retained budget. Panel (a) uses RULER Cohort A and full-set LoCoMo lexical scoring. CoMem reads its bounded pack from the full store without YaRN; the SnapKV/PyramidKV 64k/128k columns use YaRN factor 4 and see the full prompt. [†]Their native 15-cell macros include 64k/128k inputs left-truncated to the 40,960-token native window; within 8k–32k both score 99.96. Panel (b) times full-prompt prefill, compression, and 64-token decode (median of three); the final column is decode latency. These methods must prefill the full prompt before eviction. CoMem’s separate L20A cohort peaks at 18.54 GB at 128k (Appendix A.4); hardware differs, so we report no cross-table latency ratio.

trieved chunk IDs, order, and pack tokens; only replay start differs. Across 15 RULER Cohort-B cells ($n=100$ each), full replay scores 99.19 and layer-12 replay scores 96.07. The paired 3.12-point gap has bootstrap 95% CI [2.36, 3.93] and exact McNemar $p=8.79 \times 10^{-24}$ (83 full-only correct versus one layer-12-only correct). The gap is largest on distractor-heavy multikey retrieval.

For latency, each arm uses the same approximately 6.5k-token pack retrieved from 16k source contexts in three independent processes, with three warmups and 20 reads per process. Layer-12 replay is $1.403\times$ faster (931.9 to 664.4 ms), with all three process-level ratios between 1.402 and 1.404. Total-decode medians remain similar at roughly 2.76–2.86 s. A preceding component harness attributes the read reduction primarily to upper-transformer forward ($1.398\times$), with unchanged LM-head time. Retrieval, index construction, reusable store/query Write, and persistent I/O are outside this read-phase timing. The result is therefore a measured quality–latency trade-off, not quality-preserving or end-to-end acceleration.

A.5 Seed-plus-effective-batch robustness

Table 29 reports the available run-level aggregates. Across the 18 individual RULER and BABILong

Replay start	RULER	Read (ms)	p10	p90
$j=0$	99.19	931.9	931.6	942.0
$j=12$	96.07	664.4	663.8	667.1
Trade-off	−3.12 pt		$1.403\times$ speedup	

Table 31: Paired quality–latency control with the same LoRA. Quality is the 15-cell RULER macro ($n=1,500$ paired examples); the 3.12-point loss has paired-bootstrap 95% CI [2.36, 3.93]. Latency uses a fixed approximately 6.5k-token pack retrieved from 16k source contexts and reports medians across three independent processes (20 reads each). Retrieval, reusable Write, I/O, and total decode are excluded; total decode is similar across arms.

Replay pathway	RULER B	MK 8/16k	Reusable
Full replay from $j=0$	99.19	100.0	No
Continuous-prefix h_{12} oracle	99.19	100.0	No
Chunk-local cached h_{12}	96.07	92.5	Yes

Table 32: Write-side attribution with identical examples, packs, and LoRA. The continuous-prefix oracle runs layers 0–11 over the selected packed sequence for every query, then starts upper replay from that h_{12} ; it is bit-identical to full replay on all 1,500 Cohort-B examples. The deployable chunk-local path loses 3.12 points (95% CI [2.36, 3.93]; McNemar $p=8.79\times 10^{-24}$), and 7.50 points on the pooled 8k/16k multikey cells ($n=200$, CI [4.0, 11.5]). The oracle is an attribution upper bound, not a reusable cache.

Backend	Write GB/s	Fetch ms	H2D ms	Peak QPS	16M outcome
GPU resident	OOM	–	–	–	128 GiB > 95 GiB
CPU pinned	139.6	0.81	1.41	941	235 GB host
Local NVMe	2.45	13.42	0.91	264	2.18 GB host
Network/CEPH	1.66	79.20	1.20	42	2.19 GB host

Table 33: Measured persistent-store tiers at 16M tokens (128 GiB of bf16 h_{12}). Each query fetches the same 50.3 MB top-12 pack; medians use seven repetitions after two warmups, and QPS is the best of 1/4/16 concurrent clients. GPU residence fits 8M tokens (64.98 GB peak) but not 16M; off-GPU tiers retain store-size-independent transfer volume at increasing I/O latency. This microbenchmark times storage and transfer, not BM25 or model inference.

cells, the median cell-wise SD is 1.34 points and the maximum is 4.36. Because the two added runs change effective batch from 8 to 3, these numbers jointly probe initialization and optimization-noise robustness; they are not a clean three-seed variance estimate. The available evidence also does not support a multi-run claim for the exact 15-cell RULER-B, LoCoMo, or LongEval headlines.

A.6 Prompt and native-chat sensitivity

For Qwen3-8B, the GPT-4o LoCoMo judge is substantially less prompt-sensitive than token F1. CoMem scores 38.27 without chat templating and 37.76 with it; KV-Direct scores 34.59 and 38.22. In contrast, CoMem’s F1/substring accuracy changes from 9.15/23.36 to 19.51/28.65, while KV-Direct F1 changes from 9.02 to 40.06, showing that lexical metrics are strongly formatting-sensitive. We therefore retain the uniform no-chat judge comparison as the controlled headline.

For MemoryLLM, we report only the uniform no-chat row in the paper. A matched native-chat/no-chat table was not retained in the frozen manuscript; therefore we make no prompt-sensitivity claim for this external baseline and do not mix prompt protocols within an average.

A.7 Cross-scale and MoE generality

Adapter-free sweeps across six Qwen3 sizes and Qwen3-30B-A3B confirm that the split-forward implementation ports beyond the flagship. Performance is non-monotonic because split, architecture, and task differ, so we infer neither a scaling law nor a replicated quality frontier. These exploratory values and raw predictions are retained in the artifact; the Hy3 study below gives the stronger architecture-level partition check reported in this appendix.

A.7.1 Hunyuan Hy3: an 80-layer sparse MoE

We additionally port CoMem to Tencent Hunyuan Hy3 (Tencent Hunyuan Team, 2026), an 80-layer sparse Mixture-of-Experts model (192 experts, top-8 routing plus one shared expert; $\approx$ 597 GB in bf16) with a 262,144-token native window. The backbone is sharded across eight GPUs; Write and Read semantics are unchanged.

Exact partition through MoE routing. Reconstructing a contiguous forward by writing `layers[0:j]` and resuming `layers[j:80]` gives $\max|\Delta\logit| = 0.0$ at $j \in \{0, 1, 40, 80\}$, including the Dense-to-MoE boundary. The resumed half re-executes every router and expert, so the partition preserves routing exactly on this self-test.

Split depth and self-distillation. A PG19 sweep places the best readout region near $j \approx 32$ –36. At $j = 32$, self-distillation reduces the 16k multiplicative LM tax from 1.496 to 1.078 and raises top-1 agreement from 0.773 to 0.871 (Table 34).

$j=32$	ctx 8k		ctx 16k	
	gap	top1	gap	top1
zero-shot	1.378	0.823	1.496	0.773
distilled	1.071	0.895	1.078	0.871

Table 34: CoMem self-distillation on Hy3 at split $j=32$ (LoRA $r=32, \alpha=64$ on `layers[32:]`, teacher $j=0$ RAG recompute, 4000 steps of plain PG19 KL, no synthetic data). **gap** is the multiplicative LM tax (CoMem-readout perplexity / full-context perplexity; 1.0 is exact); **top1** is argmax agreement with the full forward, over matched 16-doc PG19 windows. The 16k LM tax falls from +49.6% to +7.8% and top1 rises $0.773 \rightarrow 0.871$, reproducing the 8B-backbone self-distillation effect on the 80-layer MoE.

Bounded read through 256k. With the distilled adapter and BM25 top-8, the read stays at $\approx$ 4.3–4.6k tokens from 16k through 256k. CoMem reaches 100/98 single/multikey recall at the 256k native ceiling with no OOM (Table 35).

Task	16k	32k	64k	128k	256k
niah_single_2	98	92	100	98	100
niah_multikey_1	100	94	100	100	98
read length (tok)	4.3k	4.6k	4.5k	4.4k	4.5k
context (tok)	16k	32k	65k	131k	261k

Table 35: CoMem on the Tencent Hunyuan **Hy3** backbone (hy_v3, 80-layer sparse MoE), RULER needle recall ($n=50$, official string_match_all), split $j=32$, BM25 top-8, self-distilled LoRA. The read length fed to the model stays $\approx 4.3\text{--}4.6\text{k}$ tokens while the context grows $16\times$ (16k $\rightarrow$ 256k); CoMem retains 100/98 recall at the backbone’s 256k native ceiling, with 0 out-of-memory failures. The context row reports the mean packed length of the RULER samples.

B Chat-Template-Free Statistical Verification

All statistics in this appendix were recomputed from the saved prediction shards on the shared result filesystem, without GPU reruns. The Qwen3-8B flagship uses the self-distilled $j=12$ adapter, chunk 512, top-12 retrieval, a BOS sink, and no chat template. The released adapter artifact is identified by the SHA-256 and configuration in Table 27. The main RULER directory uses the iter_bm25 implementation; the saved configuration records rounds=0 and an observed read of approximately 6.5k tokens. Every reported RULER cell contains eight of eight shards, 100 examples, no empty prediction, and zero mismatch between the stored score and a fresh official-kernel recomputation. BABILong likewise contains 100 valid examples in every cell.

B.1 RULER cohort definitions

Cohort A is the all-method table (Appendix Table 19): it substitutes niah_single_2, keeps the same multikey/tracking tasks and $n=100$, and gives CoMem 97.05. Cohort B above was freshly sampled with niah_single_3 because that is the exact task available for both YaRN arms and $j=0$; it gives 96.07. Tables mark every RULER aggregate with *A* or *B*, never subtract across them, and exclude legacy 256k cells. Table 23 contains only Cohort B.

Task	8k	16k	32k	64k	128k
niah_single_3	100.0	91.0	97.0	98.0	98.0
niah_multikey_1	94.0	91.0	99.0	90.0	93.0
variable_tracking	96.2	98.0	98.2	98.6	99.0

Table 36: RULER **Cohort B**, used for the matched length-extension and $j=0$ comparisons. The $n=100$ macro is $1441.0/15 = 96.0667$; the exact single-needle variant is niah_single_3.

B.2 LoCoMo judge protocol and denominator

Prediction shards for all methods and the adapter-free CoMem control deduplicate to all 1,986 LoCoMo items. GPT-4o judges the 1,540 answerable items in categories 1–4. Category 5 contains 446 adversarial abstention items and is scored locally: an answer is correct iff it is empty or matches the refusal rule. Thus the canonical full-set judge is over all 1,986 items, not only the API-judged subset.

Method	Cat. 1	Cat. 2	Cat. 3	Cat. 4	Cat. 5	Overall
CoMem + LoRA	26.95	19.00	30.21	69.32	2.47	38.27
CoMem frozen ($j=9$)	18.79	13.40	27.08	53.75	1.12	29.15
KV-Direct	24.11	18.69	25.00	62.19	2.69	34.59
StreamingLLM	22.70	11.21	23.96	42.21	6.95	25.63
InfLLM	18.44	14.33	25.00	33.89	7.62	22.21
MemoryLLM	15.60	8.72	15.63	27.47	0.45	16.11

Table 37: LoCoMo GPT-4o judge by category. Categories 1–4 contain 282/321/96/841 judged items; Category 5 contains 446 locally scored abstention items. Overall aggregates all 1,986 examples.

The judge was evaluated on July 22, 2026 using an OpenAI-compatible chat-completions endpoint with model name gpt-4o, seed=1, and no client-set temperature or top- p . The endpoint does not expose a dated model snapshot. Saved item-level records contain the item ID, binary parsed decision, raw response text, and model string; the anonymous artifact includes the permissible parsed decisions and their hashes rather than API credentials. Requests are retried four times with exponential backoff; unparsable responses and API failures are conservatively scored wrong. The prompt presents the question, gold answer, and prediction; it accepts any semantic match (including equivalent date phrasing), rejects contradictions, omissions, empty answers, and refusals, and requires exactly CORRECT or WRONG. The verbatim template is released with the evaluation script. Gold alternatives are joined with “OR.” A reply must begin with CORRECT or WRONG; a fall-

back substring vote is used before defaulting to wrong.

On the common GPT-4o-judged items ($n=1540$), CoMem scores 48.64 and KV-Direct 43.83. A paired per-item bootstrap of their difference (10,000 resamples, seed 1234) gives +4.81 points with 95% CI [2.34, 7.34]. Because the questions are nested within only 10 conversations, this interval is not our sole inferential check. A paired conversation-cluster bootstrap, resampling the 10 conversations rather than individual questions (same 10,000 resamples, seed 1234), gives the same +4.81 point estimate with 95% CI [1.27, 8.32] — wider than the per-item interval, as expected with only ten clusters; 8 of the 10 observed conversation-level differences favor CoMem. We do not interpret the bootstrap tail fraction as a calibrated hypothesis-test p value. We report the latter as a dependence-aware robustness check rather than claiming that 1,540 questions are independent. The canonical full-set point scores are 38.27 and 34.59, respectively.

B.3 Independent-judge audit

On July 26, 2026, we reran the binary prompt with DeepSeek-V3 on a stratified 200-item subset shared by CoMem and KV-Direct. Agreement with GPT-4o is 0.81 ($\kappa = 0.626$). Both judges preserve the ordering: GPT-4o scores 36.5/32.5 (+4.0) and DeepSeek-V3 56.0/49.0 (+7.0) for CoMem/KV-Direct. Absolute calibration differs, but the positive gap is judge-robust.

B.4 Uncertainty and aggregation checks

Equal-latency dependence reanalysis. We re-analyzed the saved anchor scores without model execution. The estimand is the equal-weight mean of the nine cell mean paired differences $\text{CoMem}(k=12)$ minus $\text{replay}(k=10)$. A *stratified fixed-cell* bootstrap keeps all nine cells and resamples exactly 100 paired examples within each cell, giving 95% intervals [−14.33, −8.78] for BM25 and [−4.56, 2.56] for BGE. A *hierarchical* bootstrap first resamples nine cell labels with replacement, then independently resamples 100 paired examples inside each selected occurrence; its intervals are [−18.67, −5.11] and [−10.67, 8.33]. Dropping one cell at a time gives ranges [−13.13, −9.50] and [−3.50, 1.75], respectively. The original pooled-IID intervals, [−14.44, −8.67] and [−4.67, 2.67], are retained only as sensitivity. All use 100,000 replicates and

deterministic base seed 20260804. The first 100 LoCoMo examples are all conversation 0, leaving one observed cluster; a conversation bootstrap is therefore not identifiable. The release includes the score-only exports (without benchmark text, answers, or predictions), script, result JSON, and SHA-256 manifests.

Per-method, 1,000-resample item-bootstrap intervals (seed 1234) are [36.20, 40.58] for CoMem’s full-set LoCoMo judge, [32.48, 36.71] for KV-Direct’s, [46.04, 50.98] and [41.23, 46.30] for their category-1–4 judge scores, and [8.60, 9.72] and [8.51, 9.61] for their F1 scores. These unpaired, item-level intervals are descriptive; the paired cluster analysis above is the appropriate dependence-aware comparison.

We do not reuse cell-bootstrap intervals from the superseded RULER cohort or legacy BAILong predictions. The separate six-length LongEval aggregate is 72.83 over 600 examples (95% CI [69.33, 76.33]); the benchmark summary uses the predefined five-length 8k–128k mean of 69.0.